

Samin Yeasar

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Research Interests

- Computer Vision and Deep Learning
- Software Engineering
- Artificial Intelligence (AI) in Interdisciplinary Research

Experience

Software Engineer

CodeMarshal IT Systems 🔗

Dhaka, Bangladesh
February 2024 – Present

- Contributed to research and development of a zero-shot **visual inspection engine** leveraging open-vocabulary detectors and vision-language models (VLMs) to automate visibility auditing.
- Contributed to the development of a **real-time monitoring system** by optimizing RTSP stream ingestion and integrating proprietary computer vision APIs for live CCTV analysis.
- Improved Bangla ALPR vehicle-class OCR accuracy from **52.54%** to **62.71%** through OCR input optimization.

Education

Khulna University of Engineering & Technology

BS in Computer Science and Engineering

March 2016 – July 2023

- GPA: 3.24/4.0 (**Average CGPA of Last 4 Semesters: 3.54**)
- **Thesis:** *A Novel DNA-based Cryptosystem for Text and Audio Data* 🔗

Publications

Climate Crisis and Food Security in Bangladesh: Utilizing Machine Learning in Geospatial Environment to Investigate Drought Tolerance of Major Crops

Preprint, 2024

- R. R. Rudra, **S. Yeasar**, T. A. Noor, et al. *Applied Random Forest and SHAP analysis to evaluate drought tolerance of five major Bangladeshi crops using 16 climatic and geographic variables.*
- DOI: [10.2139/ssrn.4974356](https://doi.org/10.2139/ssrn.4974356) 🔗 (*SSRN preprint*)

Capturing Rainfall Dynamics: The Efficacy of ARIMA Models in Environmental Forecasting

ICACE 2024

- L. C. Roy, S. K. Das, **M. S. Yeasar**, et al. *Developed an ARIMA(2,1,2)-based rainfall forecasting model using 40+ years of meteorological data from 30 stations across Bangladesh.*
- 7th International Conference on Advances in Civil Engineering, CUET, Bangladesh.

Projects

VSP-MVP: Virtual Simulated Patient Chatbot

github.com/Y3454R/vsp-mvp 

- Developed a full-stack mental health training platform implementing the **CureFun research methodology**, utilizing **FastAPI** and **LangChain** to orchestrate **GPT-4o-mini** for realistic patient simulation and automated evaluation of 8 clinical competencies.
- **Tools Used:** **Cursor (AI-assisted development)**, FastAPI, Next.js 14, LangChain, PostgreSQL, Docker

Envertor: CLI Tool

github.com/y3454r/envertor 

- Developed a simple utility CLI tool to prepare and manage .env files in CI/CD pipelines.
- **Tools Used:** Python, Regex, Click (CLI Framework).

Technical Skills

Languages: Python, JavaScript, C++

Frameworks: Pytorch, FastAPI, Next.js, Express.js

Tools: Git, Docker, Jenkins

Database: PostgreSQL

Competitive Programming

- **Intra KUET Programming Contest 2020** — Placed **30th** 

Extracurricular Activities

- **Vice-President**, Musical Association of KUET